CS 440 Assignment 1

Fast Trajectory Replanning in unknown gridworlds

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A. Replanning, head to head

expansions, trajectory and replans per world, h = manhattan, larger-g ties

	expansions	moves	replans	solved
scatter 101x101				
Repeated Forward A*	3,331	218	58	50/50
Repeated Backward A*	**24,818**	213	57	50/50
Adaptive A*	**3,265**	217	59	50/50
maze 51x51				
Repeated Forward A*	24,504	629	231	50/50
Repeated Backward A*	**59,334**	604	236	50/50
Adaptive A*	**13,797**	627	235	50/50

B. Tie-breaking (Part 3)

Repeated Forward A*, equal-f ties broken toward smaller or larger g

	smaller g	larger g	small / large
scatter	51,628	3,575	**14.4x**
maze	75,576	32,348	**2.3x**

C. Heuristics, one search, known map

expansions for a single A* search from the agent to the target

	manhattan	differential	yours
scatter	1,007	668	**436**
maze	3,675	1,089	**1,081**

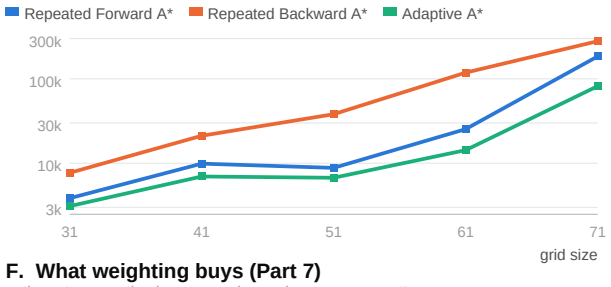
D. Landmark staleness (Part 6c)

how far the tables may drift before rebuilding, in the maze loop

rebuild after	expansions	seconds	probes
age > 1	6,306	3.19	3,435,610
age > 4	6,614	1.94	1,894,720
age > 16	7,496	0.73	680,316
age > 64	9,739	0.28	196,220
never rebuild	13,596	0.20	36,556

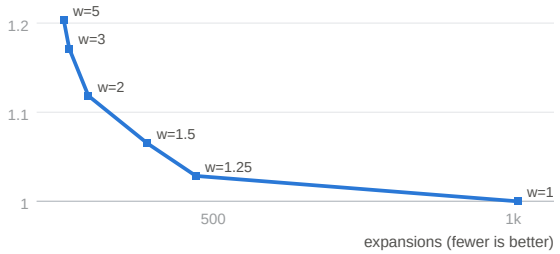
E. Cost against grid size

mean A* expansions per world, maze family, log scale

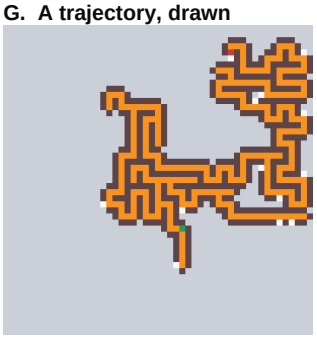


F. What weighting buys (Part 7)

path cost over optimal, one search on a known map, scatter



G. A trajectory, drawn



Adaptive A* on maze seed 753919. Orange is the route actually walked, dark cells are blockages it discovered, flat grey is terrain it never observed.

Q1 Why east, and why Manhattan is consistent

Since the agent doesn't know of the walls, and east is the closest square to the target given the choices of west north east, the agent chooses east. Even under Manhattan distance, the answer is the same since the agent can only move in the cardinal directions anyways. Via the inequality: c(s, s') = 1 in this grid, and h(s) - h(s') <= 1 is always true since the Manhattan distance changes by 1 at most. Hence h(s) <= h(s') + 1 = h(s') + c(s, s'). And h(target) = 0 which is consistent.

Q2 Termination, and the bound on the number of moves

Since the agent explores until it hits a wall, and there are N cells, the agent can "find" walls at most N times, so we can at most have N replans. Then, since there are N cells, and A* never visits the same cell twice, each plan can have at most N steps. Hence, even in the worst case, we have N plans with N steps each, so the agent must find the target (or show there is no path at all) in finit time, which is bounded by N * N = N^2.

Q3 Tie-breaking: what you measured and why it happens

Breaking ties with a larger g made the number of expansions less, since we are basically doing DFS and traversing one path all the way. We see more of an effect in scatter vs maze since in scatter the open space is where the paths tie in distance, vs in maze the constricted paths have different lengths so we have less ties overall.

Q4 Forward vs. backward: where the asymmetry comes from

Since the blockages were near the start, forward A* finds them early then pivots, so the cost of replanning is lower. Backward A* makes it all the way to the start then has to replan, so the sunk cost of the plan it just made is higher. Between replans, forward A* moves the start each time, while backward A* keeps it fixed.

Q5 Adaptive A*, and whether the difference is real

The updated h values are admissible because h_new(s)=g(goal)-g(s) does not overestimate. They are consistent since neighboring g values differ by at most the move cost. If costs increase, old h values stay safe. H0: Adaptive and Forward have no difference in expansions. Worlds are paired because they use the same maps. Signed-rank assumes symmetric paired differences. For both families, a small p rejects H0; p shows significance, not effect size. Failure to reject means insufficient evidence of a difference.

Q6 Your differential heuristic, and when it fails to pay

I picked passable corners, edge midpoints, and interior points so the landmarks were spread out. |d(l,s)-d(l,goal)| is admissible by triangle inequality and consistent since one move changes it by at most the move cost. Older tables had fewer known walls, so their distances can only be too small and stay admissible. We use max because each value is a lower bound, while summing can overestimate. Landmarks helped scatter more than mazes; rebuilding too often adds enough overhead to cancel the gain.

Q7 Weighted A*: the bound, and why the loop erases the gain

For Weighted A*, when the goal is popped its weighted f is no larger than an open node on an optimal path. Since g on that node is no more than its optimal cost so far and h is admissible, the goal cost is at most w times optimal. Larger w usually means fewer expansions but a worse path, and replanning can repeat that tradeoff. Adaptive A* cannot safely learn C-g(s) from weighted search because C may already be suboptimal, so the learned h can overestimate.